Xilinx FPGA boards for beginners

As a FPGA website for beginners or students, I always look for good and cheap Xilinx FPGA boards for beginners. There are many cheap Xilinx FPGA boards, but many of them are not easy to use especially for students or beginners; they do not offer onboard 7-segment LEDs, switches, LCD, RS232/ VGA port, other needed peripherals for beginners playing around with the board. Many FPGA boards from Xilinx are very user-friendly and they provide many onboard devices, but the price is not affordable for beginners.

Today, I will present my most recommended and affordable Xilinx FPGA development boards for beginners or students. These FPGA boards are affordable and offer all beginners need for their practice and academic projects.

1. Elbert V2 - Xilinx Spartan 3A FPGA Board (29.95USD)

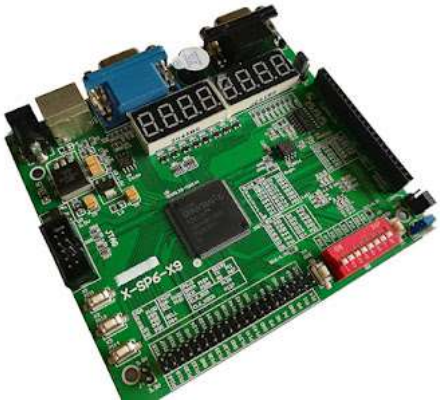
The reasons why this Xilinx Spartan 3A FPGA Board is an excellent choice and recommended for beginners are as follows:

- On board Xilinx FPGA Spartan 3A: XC3S50A-TQG144
- Only 29.95USD: affordable FPGA Board for beginners
- 8 single LEDs, 6 buttons, 8 Switches, 3 7-segment LEDs, other 39 IOs
- On board Audio Codec, VGA port, Micro SD card
- Good English documentation and sample code for the FPGA board
- Good feedback and reviewed by many students
- Another good point is the FPGA board supplier provides a configuration tool to program the FPGA board via JTAG/ USB programming cable. Thus, you can program the Xilinx FPGA board using normal Laptops or PCs. Of course, you need to use Xilinx ISE for generating the programming bitstream file.

The supplier provides a very good quality FPGA board with many peripherals for playing and also offers a very low price. Another important thing is the FPGA board supports FPGA configuration via JTAG/ USB programming cable. Personally, I think this FPGA board is the best choice for beginners.

If the supplier of this FPGA board can add more 7-segment LEDs, RS232 port for serial communications or onboard LCD for text displaying, the FPGA board can be even more and more attractive to beginners.

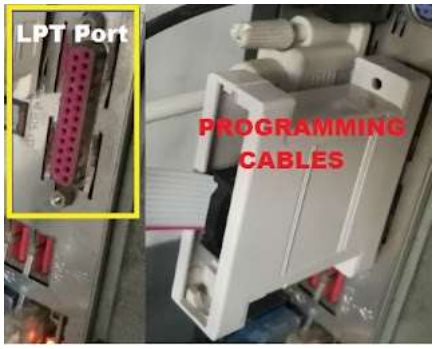
2. Xilinx Spartan 6 FPGA Board From AliExpress (23.66USD)



Following are the very good points from the Spartan 6 FPGA Board to make this FPGA board recommended and affordable for beginners:

- **On board Xilinx FPGA Spartan 6: XC6SLX9-TQG144**
- **Only 23.66USD: a bit cheaper FPGA Board vs the above FPGA board, an affordable FPGA board for beginners**
- **12 single LEDs, 8 7-segment LEDs, Switches, other IOs which are very good for beginners to try simple projects**
- **On board I2C, VGA, UART port for serial communications**
- **On board PS/2 keyboard port**
- **Offer both Verilog/ VHDL sample code, bit file to test the FPGA board**
- **Good English documentation for the FPGA board**
- **Good feedback and reviewed by many students**

The only disadvantage of these kinds of custom FPGA boards is that it is not supported by Xilinx ISE to download the programming file to the FPGA board. The programming file may be generated by Xilinx ISE, but it will need another downloader to download the programming file to the FPGA board. Unfortunately, the downloader of this FPGA board does not support FPGA programming via JTAG/USB cable and they use a parallel downloader cable (JTAG + the old LPT Port) as shown in the following picture. Students must have a PC with the LPT port to download the programming file to the FPGA board.

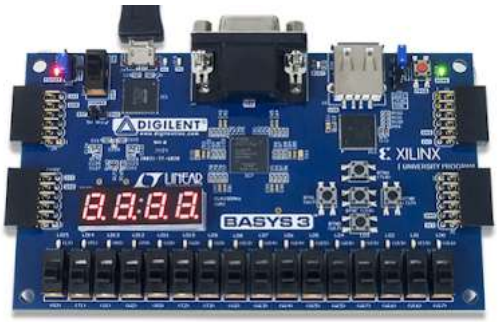


3. Mimas V2 Xilinx Spartan 6 FPGA Board 49.95USD

This FPGA board is also provided by the same FPGA board supplier as the Elbert V2 FPGA board, so they also offer the following good features for the FPGA board:

- On board Xilinx FPGA Spartan 6: XC6SLX9-CSG324
- Only 49.95USD: a bit higher price than the Elbert V2 version, but it is still an affordable FPGA board for beginners.
- The difference is the onboard 512Mb DDR SDRAM, and the other features are similar to the Elber V2 FPGA board.
- 8 single LEDs, 6 buttons, 8 Switches, 3 7-segment LEDs, other 32 user-defined IOs.
- On board Audio Codec, VGA port, Micro SD card
- JTAG/ USB programming cable for FPGA programming
- Good documentation, sample code for testing the FPGA board and good feedback from students

4. Xilinx Basys 3 Artix-7 FPGA Board for beginners 79USD



The Xilinx Basys 3 FPGA Board is of course perfectly designed and suited for beginners or students. That's why the FPGA board is very popular in university's labs for students courses. The Xilinx FPGA board is also designed for the latest design suite Xilinx Vivado (Free Webpack Version available). Following are the good features of the recommended and affordable Xilinx Basys 3 FPGA board:

- Xilinx Artix-7 FPGA: XC7A35T-1CPG236C
- 79$ affordable if you can get academic pricing as a student, 149$ for normal price
- 16 single LEDs, 16 switches, 5 buttons, 4 7-segment LEDs
- USB-UART bridge for serial communications
- VGA port, onboard XADC, 4 PMOD ports
- USB host for keyboard/ mice
- Of course, students will never worry about programming cable. The FPGA board offers JTAG-USB port for FPGA programming.

The only thing that beginners may feel uncomfortable is the price of the FPGA board. However, you can't expect a lower price with this perfect FPGA board for beginners. The FPGA board offers way better onboard I/O devices and all needed FPGA support circuits; it is super easy to use and provided by a reliable FPGA board supplier. Thus, this Xilinx FPGA board is still a very good choice for students or beginners.

Above are my four recommended, but affordable Xilinx FPGA boards for beginners or students. The Xilinx FPGA boards provide many I/O devices and supporting circuits for student's practice and importantly the FPGA boards price is affordable for beginners.

Recommended and affordable Altera FPGA boards for students

What is FPGA Programming? FPGA vs Software Programming

Recommended FPGA projects for beginners:

1. What is an FPGA? How VHDL/Verilog works on FPGA
2. How to load a text file into FPGA using VHDL/ Verilog
3. 8-bit Microcontroller Design on FPGA using VHDL
4. Matrix Multiplication on Xilinx FPGA
5. Digital alarm clock on FPGA
6. Cryptographic Coprocessor Design on FPGA
7. 16-bit single-cycle MIPS processor design on FPGA
8. Microcontroller implementation on FPGA
9. Car Parking System on FPGA

10. Image processing on FPGA using Verilog HDL
11. Traffic Light Controller on FPGA
12. Alarm Clock on FPGA
13. Comparator design on FPGA using Verilog
14. 16-bit RISC Processor design on FPGA
15. Pipelined MIPS Processor Design on FPGA
16. Basys 3 FPGA OV7670 Camera

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